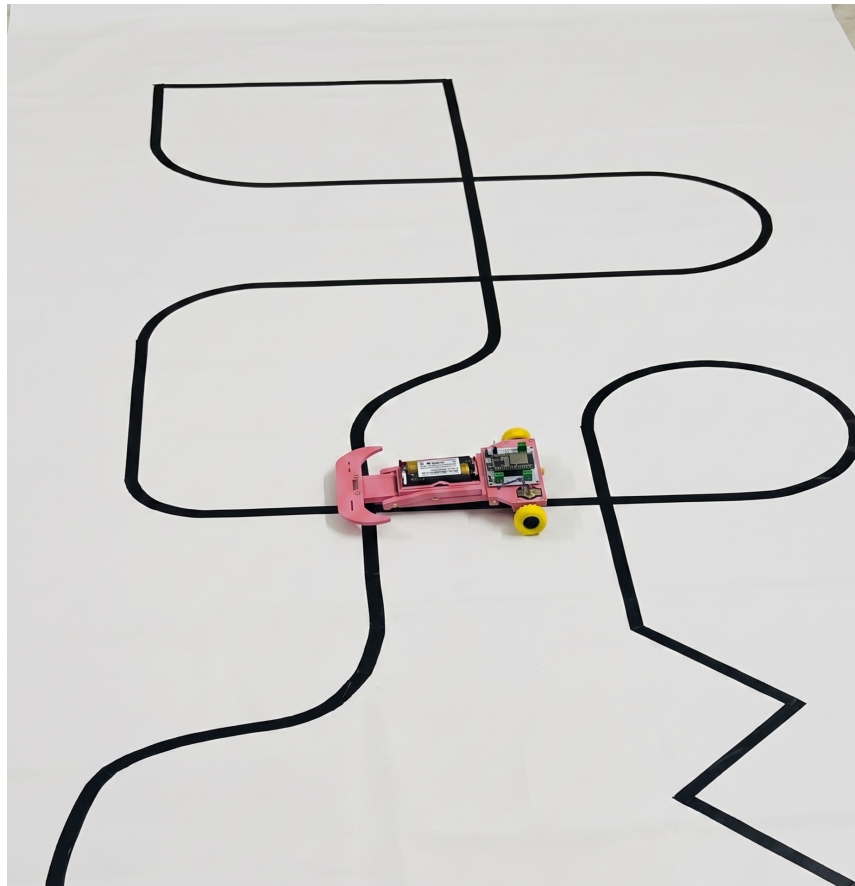


Zeroldea

Build Log — Custom PID Line Follower Robot

A pictorial record of the full build, from the first concept sketch to the finished robot running on the test track — chassis design, custom PCB fabrication, and the reasoning behind the key hardware decisions.



Built for Trailblaze — Cognizance, IIT Roorkee

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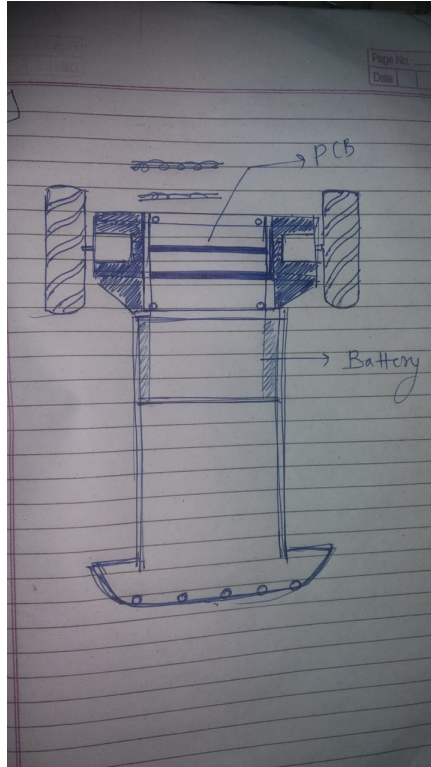
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STAGE 1

Concept Sketch

The project started on paper, not in CAD. The layout decided here — two drive motors at the rear, a battery bay running down the spine of the chassis, and the custom PCB mounted directly above the motors — carried through, largely unchanged, to the final robot.

The core design intent was to keep the electronics low and central for a stable center of mass, while leaving the rear open for the wheels and the front free for a sensor arm to be added later.



Very first prototype sketch — motor and PCB placement, and battery bay, planned before any CAD work began.

STAGE 2

Cardboard Prototype

Before committing to 3D printing, the layout was validated physically. A cardboard cutout stood in for the chassis, with the ESP32 dev board, motor driver, and battery temporarily wired on top using jumper leads.

This step caught real placement issues early — how much clearance the wheels needed from the body, where the wiring harness could realistically be routed without fouling the wheels, and roughly how much front overhang a sensor array would need — all before spending any time or filament on a 3D-printed version.



First cardboard prototype, built for testing layout and wiring before finalizing the CAD chassis.

Motor Driver Decision — TB6612FNG vs. L298N

The L298N is the default motor driver in almost every beginner line-follower tutorial, so before finalizing the PCB schematic, a direct technical comparison was done to check whether it was actually the right part for this robot.

Parameter	L298N	TB6612FNG
Switching element	BJT	MOSFET
Voltage drop	~1.4 V	~0.5 Ω on-resistance
Max motor voltage	46 V	13.5 V
Logic voltage range	4.5 – 7 V	2.7 – 5.5 V
Standby current	N/A	~0.1 μ A (STBY pin)
Thermal shutdown	Not implemented	Hardware-automated
Brake mode	Software-simulated	Hardware short-brake

For a small, battery-powered, low-voltage robot, the L298N's BJT-based H-bridge wastes a meaningful fraction of the supply voltage as heat — it is simply built for higher-voltage, higher-current loads than a line follower needs. The TB6612FNG's MOSFET stage has far lower conduction loss at these voltages, supports a genuine hardware brake (useful given how often a PID line follower reverses direction), and exposes a standby pin for clean, low-power shutdown.

The one caveat: unlike most L298N Arduino libraries, the TB6612FNG requires explicit standby-pin management — it will not drive the motors unless STBY is held HIGH. This is handled directly in the firmware's `setup()` routine.

Custom PCB — Schematic & Fabrication

All electronics — the ESP32-DevKit-V1, the TB6612FNG motor driver, an LM7805 voltage regulator, and the IR sensor header — were laid out on a single custom PCB in KiCad, silkscreened "ZEROIDEA". This replaced what would otherwise have been a breadboard and a mess of jumper wires with a compact, reliable board.

The board was fabricated through JLCPCB as 5 pieces with basic assembly, for a total order cost of \$42.78 including shipping — a small price for the reliability and space savings gained over a hand-wired protoboard.

The screenshot shows the JLCPCB Order History page. The top navigation bar includes the JLCPCB logo, a download link for JLCONE, a search icon, a shopping cart icon with a red notification bubble, an 'Order Now' button, and a 'My Account' dropdown. The left sidebar contains links for 'Order History', 'PCB/PCBA/Stencil', 'Flex Heater', '3D', 'CNC/Sheet Metal', 'JLCMC', 'PCB Layout', 'PCBA (Parts Selection)', 'Projects', 'Parts Manager', 'Payments', and 'support' (Account Manager). The main content area is titled 'All Orders' and includes filters for 'Order Type', 'Date', and 'Order #, Gerber file name...'. Below this, there's a summary of the selected order (2025-02-22 17:45:43, W2025022220151024, PCB/PCBA) with a table showing 'Product Details', 'Price', 'All Status', and 'Operation'. The summary includes a 'Shipped' status, 'Air Registered Mail', 'Shipment Tracking', and buttons for 'Reorder', 'Order Details', and 'Invoice'. The detailed list below shows two items: a 'PCB Prototype' (Order #: Y6-9404558A, 5pcs, \$2.00) and an 'Economic PCBA' (Order #: SMT025022260598-940..., 5pcs, \$30.81). Both items have links for 'Product Details', 'Production Progress', and 'Quality Complaint'.

Product Details	Price	All Status	Operation
PCB Prototype Order #: Y6-9404558A Build Time: 24 hours Product Details	\$2.00 5pcs	Line_Follower_PID_Y6 Production file ↓ ● Production Progress	Quality Complaint
Economic PCBA Order #: SMT025022260598-940... Build Time: 2 - 3 days Product Details	\$30.81 5pcs	bom.csv positions.csv ● Production Progress	DFM Analysis Quality Complaint

JLCPCB order history — 5x PCB prototype + 5x economic PCBA, ordered as "Line_Follower_PID_Y6".



The fabricated boards arriving from JLCPCB.



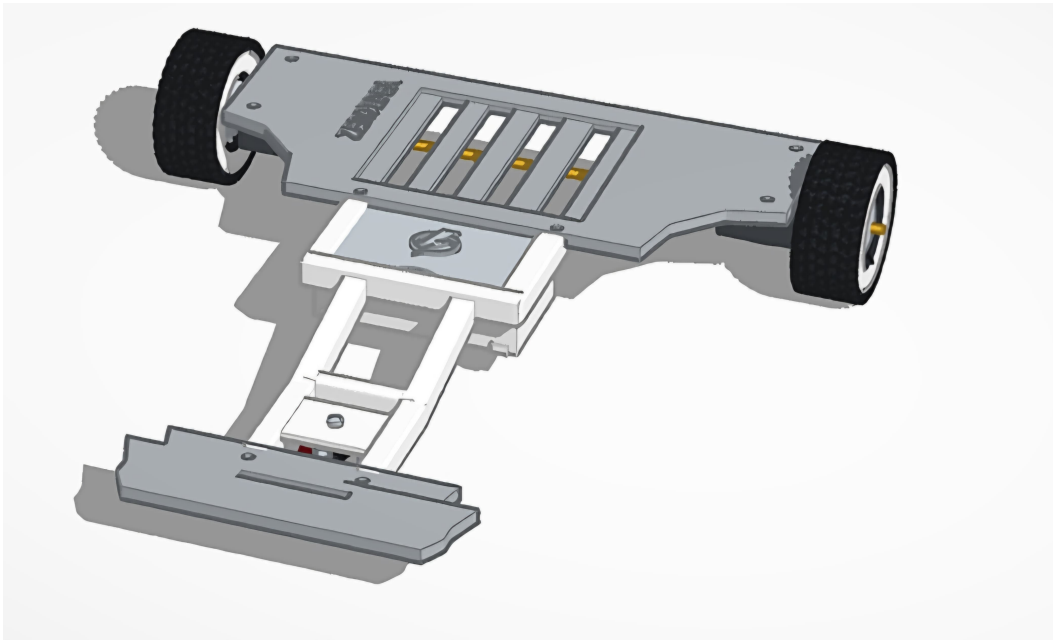
The finished, assembled ZEROIDEA board — ESP32-WROOM-32, TB6612FNG, and screw terminals for motors and battery.

STAGE 5

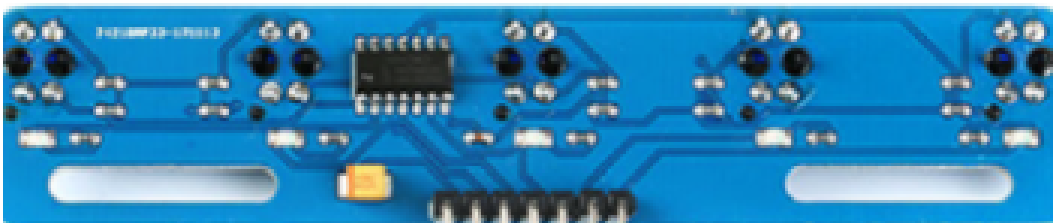
Chassis CAD & 3D Printing

With the electronics layout locked in from the cardboard prototype, the chassis was modeled in CAD and iterated through several revisions — a base body, dedicated left/right motor mounts, a caster-ball holder for the third support point, a battery bay with a removable lid, and a cantilevered front arm to carry the IR sensor array as far forward as practical (maximizing look-ahead distance for the PID loop).

The IR sensor array itself was designed with reference to standard 5-channel analog reflectance boards, adapted to mount directly at the tip of the printed arm.



CAD render of the full chassis assembly — motor mounts and wheels, main body with battery bay, and the extended sensor arm.



Reference 5-channel IR reflectance sensor board used to inform the sensor_array CAD design.

STAGE 6

Assembly

The 3D-printed chassis (printed in pink PLA) was assembled with the finished ZEROIDEA PCB mounted flat on top, directly above the motors — matching the original concept sketch. Wheels, motor wiring, and the battery bay were fitted next.

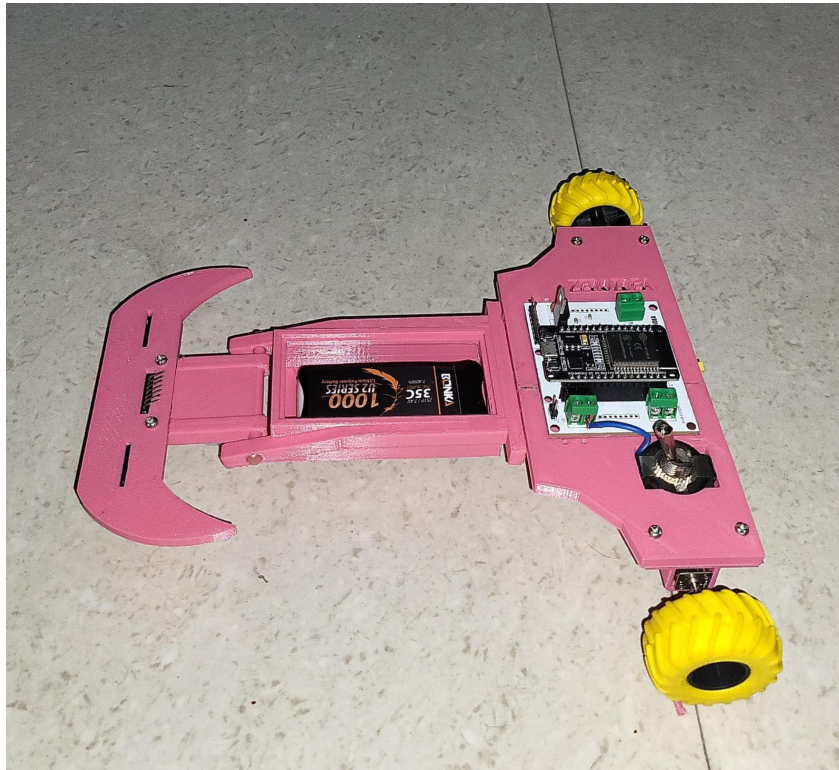


Chassis and PCB assembly complete — ESP32 board and terminal blocks visible, mounted on the printed body.

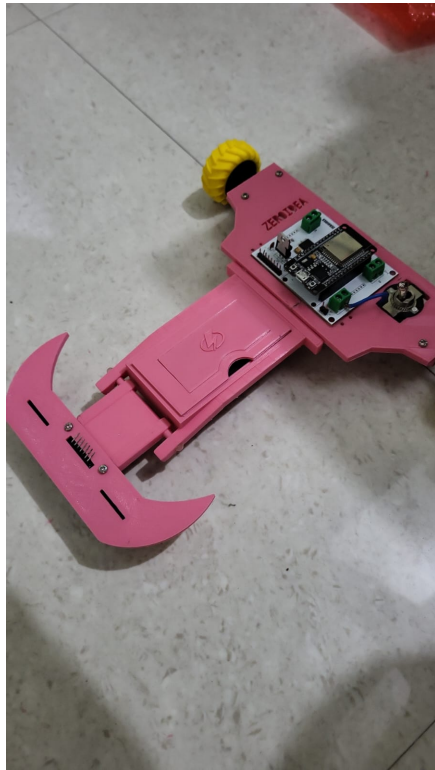
STAGE 7

Final Build

The completed robot: a 2S Li-Po (7.4 V, 1000 mAh, 35C) seated in the open-slide battery bay, the ZEROIDEA PCB and ESP32 mounted centrally, and the 5-channel IR array at the tip of the extended front arm, positioned so the caster ball trails just behind the sensor's read point on the line.



Full final assembly — battery installed, sensor arm extended, wiring complete.

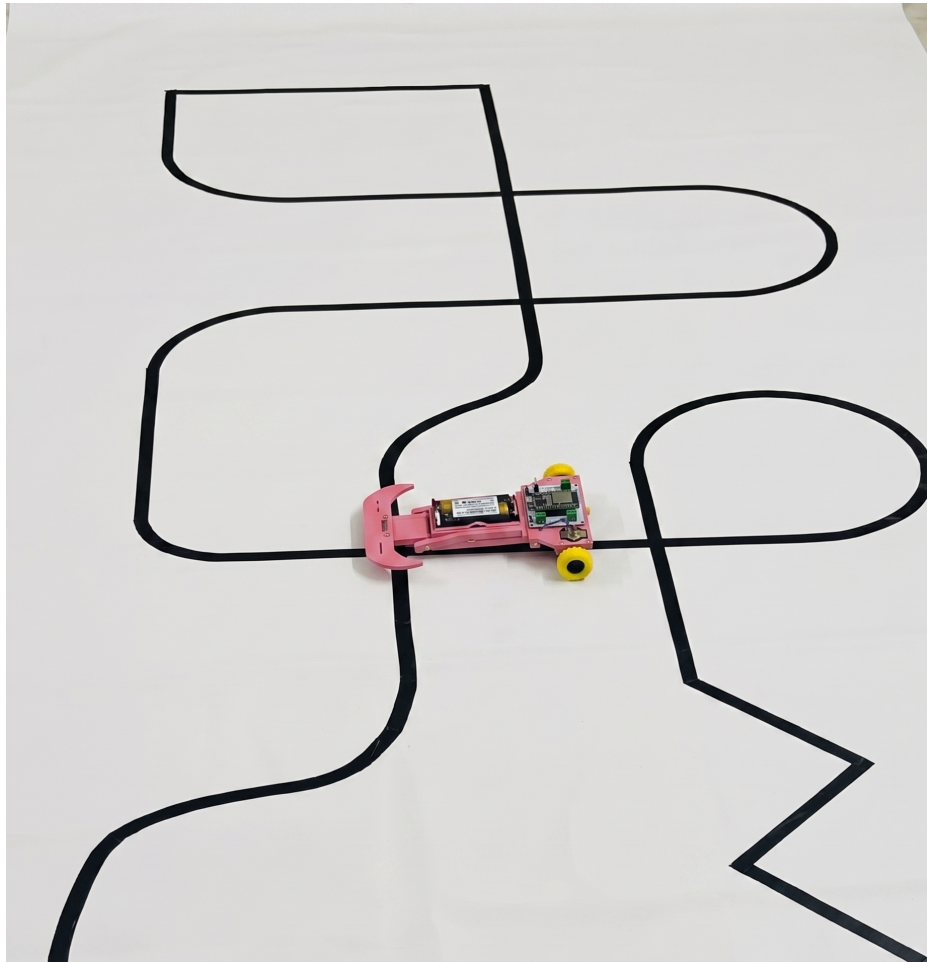


Full assembly, alternate angle.

STAGE 8

On the Test Track

The finished robot was tested on a hand-built serpentine track combining S-curves, loops, and sharp zig-zags — deliberately harder than the Trailblaze qualifying map, to stress-test the PID gains found through Bluetooth live-tuning. The robot held the line through the full course at approximately 2 m/s.



Zeroldea navigating the test track — S-curves, loops, and sharp direction changes.